

Renewable Energy Adoption: A 2024 Sector Review

This report summarizes global renewable energy adoption trends between 2019 and 2024, with a focus on solar and wind capacity growth, regional deployment differences, and policy drivers behind the acceleration observed in the final two years of the study period.

1. Overview

Renewable energy capacity has grown substantially faster than forecast in most major markets since 2022. Two factors are most frequently cited by analysts: a sharp decline in solar panel manufacturing costs driven by expanded production in East Asia, and a wave of national subsidy programs introduced following the 2022 global energy price shock. Wind capacity grew more steadily across the same period, constrained less by cost and more by permitting timelines for offshore installations.

The chart below illustrates the divergence between solar and wind growth rates over the six-year study window. Solar capacity, which trailed wind for most of the period, overtook it in 2024 for the first time.

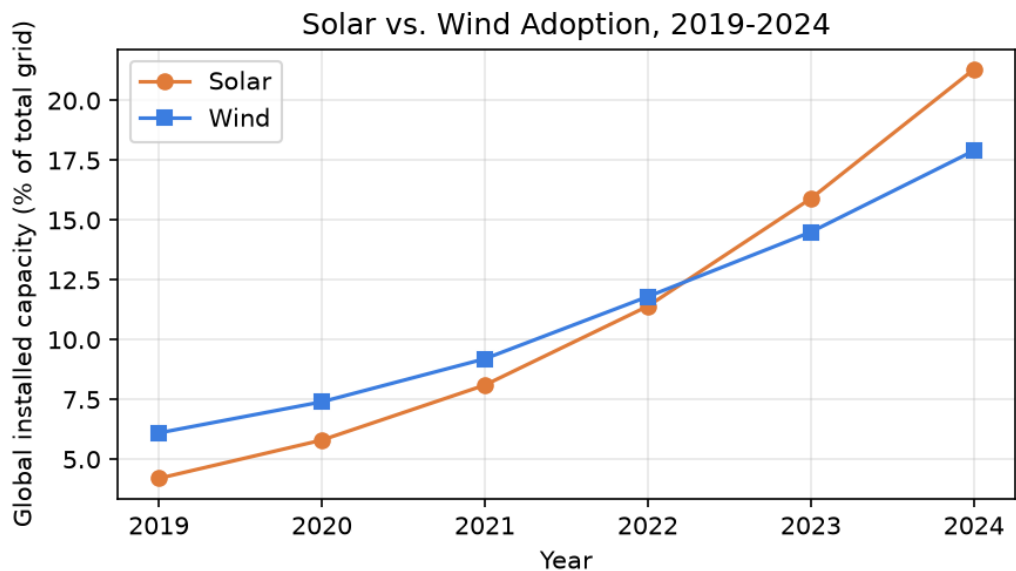


Figure 1: Global installed solar and wind capacity as a percentage of total grid capacity, 2019-2024. Solar overtakes wind for the first time in 2024.

2. Regional Adoption Rates

Adoption rates vary significantly by region, driven largely by differences in national subsidy structures and grid infrastructure readiness. The table below reports each region's renewable share of total electricity generation for 2023 and 2024, along with the year-over-year percentage point change.

Region	2023 Renewable Share	2024 Renewable Share	YoY Change
Northern Europe	58.2%	64.7%	+6.5 pp
East Asia	31.4%	39.1%	+7.7 pp
North America	27.8%	32.0%	+4.2 pp
South Asia	19.6%	24.3%	+4.7 pp
Sub-Saharan Africa	14.1%	16.8%	+2.7 pp
Oceania	35.0%	41.5%	+6.5 pp

East Asia recorded the largest year-over-year gain at 7.7 percentage points, attributed primarily to accelerated solar deployment. Sub-Saharan Africa recorded the smallest gain, constrained by limited grid infrastructure investment relative to installed generation capacity.

3. Policy Drivers

Three policy mechanisms appear most strongly correlated with above-average regional adoption growth in this study: direct capital subsidies for residential and utility-scale solar installation, streamlined permitting for offshore wind projects, and carbon pricing schemes that raise the relative cost of fossil generation.

Northern Europe's leading position is attributed largely to a combination of all three mechanisms operating simultaneously since 2021, alongside grid infrastructure that was already well-suited to variable renewable input before the acceleration began. East Asia's rapid 2024 gains are attributed primarily to manufacturing-driven cost declines rather than policy incentives, since several of the region's largest markets introduced no new subsidy programs during the study period.

4. Outlook

Analysts project continued acceleration through 2026, with solar capacity growth expected to outpace wind for the remainder of the decade due to shorter installation timelines and continued cost declines. Offshore wind is expected to see renewed growth beginning in 2026 as a wave of projects currently in permitting reach construction.